

Quantum Gravitation

Problem Set 4

Yuran Seo

Due 2026-09-27 23:59

Topic

Massive Spin-Two Representations

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Massive Spin-Two Representations. State all assumptions and conventions.

$$h_{\mu\nu} = h_{\nu\mu}, h_{\mu}^{\mu} = 0$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$k^{\mu} h_{\mu\nu} = 0$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$\sum_{s=1}^5 e_{\mu\nu}^s e_{\rho\sigma}^{s*} = \Pi_{\mu\nu,\rho\sigma}$$

Show enough intermediate work that another student can reproduce the calculation.